

Mean Splitting is Hard to Beat for Univariate Distributional Forests

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Abstract

Distributional random forests replace mean-based CART splitting with distributional split criteria, such as maximum mean discrepancy, in order to adapt forest weights to the full conditional law of a response. In univariate tabular regression, however, the extra sensitivity is not automatically useful: most finite samples appear to be dominated by location signal, while higher-order distributional structure is weak or too noisy to estimate at split time. Across synthetic diagnostics, oracle comparisons, adaptive-frequency variants, multivariate probes, and real-data benchmarks, we find that CART-style splitting is the stronger default for univariate distributional prediction. The evidence supports a simple characterization: distributional splitting helps only when non-location structure is both present and estimable; otherwise it dilutes split-selection power away from the mean.

1 Question

Random forests for distributional prediction estimate not just a conditional mean, but a conditional law. Quantile regression forests do this with standard CART tree structure and distributional leaf weights [2]. Distributional random forests instead propose splitting rules that compare the full response distributions in candidate children [1]. This is a natural generalization, especially for multivariate responses where a scalar ordering or separate marginal quantile fits are unsatisfactory.

The narrower question studied here is deliberately less ambitious: *for univariate tabular regression, is distributional splitting a better default than CART-style mean splitting?* This note records a negative answer with a positive interpretation. CART-style splitting is difficult to beat in this regime, not because distributional criteria are incoherent, but because the finite-sample split-selection problem is usually mean dominated.

2 Split Criteria as Signal Allocation

For a node with observations (X_i, Y_i) , a candidate split $A \mid A^c$ is chosen by maximizing an empirical separation score between the child responses. CART uses the between-group mean separation

$$\frac{n_L n_R}{n^2} \|\bar{Y}_L - \bar{Y}_R\|^2,$$

which is the identity-kernel mean-embedding score. Random Fourier feature MMD splitting replaces Y by a complex feature map $\psi(Y)$ and scores

$$\frac{n_L n_R}{n^2} \left\| \frac{1}{n_L} \sum_{i \in L} \psi(Y_i) - \frac{1}{n_R} \sum_{i \in R} \psi(Y_i) \right\|^2.$$

With enough features and enough data this score can detect changes beyond the mean. At finite sample sizes it also spends split-selection capacity on noisy higher-order coordinates.

This suggests the working characterization tested below. CART concentrates on the dominant location signal. MMD spreads sensitivity across location and higher-order distributional structure. MMD should win only when the additional structure is strong enough, or the sample size large enough, for its split-time signal-to-noise ratio to exceed the cost of estimating it.

3 Evidence

The experiments were run as paired-seed comparisons: the train/test split, forest seeds, tree hyperparameters, honesty setting, and cutpoint grid were held fixed while varying only the split criterion. Metrics are RMSE for the conditional mean and CRPS / energy score for distributional prediction.

Synthetic diagnostics. The first set of experiments swept the univariate synthetic mechanisms used in the DRF literature, plus targeted variants designed to isolate mean shifts, scale shifts, quantile changes, and weak higher-moment effects. CART dominated or tied MMD in the regimes where the conditional law was primarily location driven. MMD became plausible only when higher-order structure was deliberately made large enough to be visible at split time.

Table 1: Synthetic univariate summary. Replace this placeholder with paired mean differences, e.g. MMD minus CART, so negative CRPS/energy indicates an MMD improvement.

dataset	honesty	Δ RMSE	Δ CRPS	Δ energy
paper_quantile_1	0.5	TODO	TODO	TODO
paper_quantile_2	0.5	TODO	TODO	TODO
paper_quantile_3	0.5	TODO	TODO	TODO

Adaptive and oracle checks. Adaptive-frequency MMD was used as a rescue attempt: if uniform RFF averaging was hiding useful distributional signal, selecting the strongest coordinates at each split should expose it. Oracle and adaptive runs did not change the main conclusion. This points away from a simple tuning failure and toward the finite-sample allocation explanation.

Real data. On univariate tabular benchmarks, CART-style splitting was again the stronger default. The exception worth isolating is California housing, where MMD can look competitive. That case confounds two explanations: the dataset is large, and housing prices may contain estimable heteroskedastic structure. A subsample experiment over $n \in \{2000, 4000, 8000, 16000\}$ separates these readings. If MMD improves only as n grows, the evidence favors the estimability threshold story; if it improves already at small n , the evidence favors unusually strong distributional structure.

4 Characterization

The empirical pattern can be summarized as follows.

For univariate distributional forests, distributional splitting helps when non-location conditional structure is present and estimable at node scale. In mean-dominated finite

Table 2: Real-data paired-seed summary. Fill with means and standard errors for CART and MMD, or paired MMD–CART differences.

dataset	n_{train}	honesty	criterion	CRPS	fit time
diabetes	TODO	0.5	CART	TODO	TODO
diabetes	TODO	0.5	MMD	TODO	TODO
california	TODO	0.5	CART	TODO	TODO
california	TODO	0.5	MMD	TODO	TODO

samples, CART-style splitting is the better default because it allocates split-selection power to the strongest signal.

This statement is intentionally scoped. It is not a claim that distributional forests are inferior to quantile regression forests, nor that MMD is a poor criterion in general. The core motivation for DRF is multivariate response distributions, where CART/QRF-style univariate orderings are not a complete solution. The claim here is about the common univariate tabular setting: when the target is scalar and the conditional law is mostly a shifted version of itself, a distributional split criterion is often paying variance to chase weak structure.

5 Implications

For practice, the recommendation is simple. Use CART-style splitting as the univariate default, then use distributional criteria as targeted tools for data sets with clear heteroskedasticity, tail behavior, or non-location structure. When distributional splitting is used, paired-seed comparisons and honesty settings should be reported: in these experiments, honesty can be as important as the criterion itself.

For research, the next useful theory is not a generic dominance theorem. The useful object is a finite-sample comparison of split-score signal-to-noise: mean separation versus kernel mean separation under local alternatives. Such a calculation would turn the empirical characterization above into a sharp threshold statement.

6 Limitations

The experiments here are mostly univariate. They do not exhaust the design space where DRF is most naturally motivated: multivariate responses, dependence structure, and conditional distributional changes with little or no mean shift. They also compare CART-style splitting to MMD splitting inside the same forest implementation; a full external QRF package comparison is a separate baseline.

References

- [1] Domagoj Čevič, Loris Michel, Jeffrey Näf, Nicolai Meinshausen, and Peter Bühlmann. Distributional random forests: Heterogeneity adjustment and multivariate distributional regression. *Journal of Machine Learning Research*, 23(333):1–79, 2022.
- [2] Nicolai Meinshausen. Quantile regression forests. *Journal of Machine Learning Research*, 7: 983–999, 2006.